

ARCHITECTURE DECISION RECORD

ADR-002: Program Identity and Naming Hierarchy

Governed Clinical Planning Language (GCPL)

Preferred descriptive title: Governed Clinical Language for AI-Assisted Radiation Treatment Planning

Subtitle: A Model-Based, Safety-Constrained Orchestration Framework

Implementation / working name: Natural-Language Treatment Planning System (NL-TPS)

Decision version: 0.1

Date: 19 August 2026

Status: Provisionally accepted for controlled non-clinical program use; named institutional approval not recorded

STATUS AND SAFETY NOTICE

This decision changes program identity and terminology, not intended use, requirements, safety controls, professional authority, regulatory status, commissioning status, or clinical authorization. Existing NL-TPS identifiers, filenames, namespaces, and trace links remain valid.

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Document control

Field	Controlled value
Decision owner	Program sponsor and architecture board - proposed; named accountable owner requires governance recording
Decision scope	Program title, technology name, implementation alias, publication title, terminology interpretation, and controlled propagation
Excluded scope	Intended use, regulatory claims, requirement meaning, clinical workflow authority, release authorization, and repository renaming
Related decisions	ADR-001; ENG-PKG-01 AB-001 through AB-012
Supersedes	No controlled requirement or identifier; prospectively supersedes use of “Natural-Language Treatment Planning System” as the only description of the program
Recorded approvals	None in this document; provisional adoption is not evidence of institutional, regulatory, clinical, or publication approval

Decision summary

The program shall use **Governed Clinical Language for AI-Assisted Radiation Treatment Planning** as its preferred full descriptive title and **A Model-Based, Safety-Constrained Orchestration Framework** as its subtitle.

The underlying technology shall be named **Governed Clinical Planning Language (GCPL)**. The current software implementation, repository, and historical product identifier shall remain **Natural-Language Treatment Planning System (NL-TPS)**. A concise implementation reference is therefore “the NL-TPS implementation of GCPL.”

The research and formal R&D title shall be:

Design and Realization of a Governed Clinical Planning Language for AI-Assisted Radiotherapy: A Model-Based Framework for Intent Compilation, Deterministic Safety, Traceability, and Commissioned Execution

This hierarchy recognizes that natural language is an input modality, while the principal contribution is a governed, typed, machine-verifiable clinical language operating through deterministic safety controls and commissioned services.

1 Context

The original NL-TPS name correctly described the initiating concept but can now be misread as either a chatbot added to a treatment planning system or an autonomous replacement TPS. The controlled architecture is materially different. It is a command, reasoning, collaboration, and execution-control layer over validated radiotherapy systems. Human or spoken language is compiled into typed intent; a deterministic kernel evaluates authority, state, evidence, context, and commissioned scope; governed adapters invoke approved services; accountable professionals

retain decision authority; and external systems retain calculation and record authority.

The name must therefore expose the safety and systems-engineering contribution without discarding the identifiers already embedded across requirements, interfaces, risks, tests, component allocations, scripts, namespaces, and repositories.

2 Controlled naming hierarchy

Level	Controlled name	Required use
Preferred descriptive title	Governed Clinical Language for AI-Assisted Radiation Treatment Planning	Program covers, executive summaries, external descriptions, and future controlled documents where the full descriptive title is appropriate
Subtitle	A Model-Based, Safety-Constrained Orchestration Framework	Accompanies the preferred descriptive title; describes architecture without implying autonomous clinical authority
Technology / method	Governed Clinical Planning Language (GCPL)	Names the typed semantic language, governance model, safety policy, release profiles, and orchestration method
Implementation / product	Natural-Language Treatment Planning System (NL-TPS)	Names the current implementation, repository, service family, historical controlled corpus, and trace-compatible product identifier
Combined reference	GCPL / NL-TPS or “the NL-TPS implementation of GCPL”	Used during transition and whenever technology and implementation must be distinguished
Research title	Design and Realization of a Governed Clinical Planning Language for AI-Assisted Radiotherapy: A Model-Based Framework for Intent Compilation, Deterministic Safety, Traceability, and Commissioned Execution	Dissertation, formal R&D program, or paper-series title, subject to the applicable academic or publication approval

GCPL is the acronym for *Governed Clinical Planning Language*. It is not presented as the initialism of every word in the longer descriptive title. Authors shall not introduce a competing acronym for the descriptive title.

3 Meaning of the title

Term	Controlled interpretation
Governed	Requirements, evidence, authority, risks, release profiles, commissioned-use envelopes, models, interfaces, changes, and approvals are controlled and versioned rather than inferred from conversational text at runtime.
Clinical language	Clinical intent is represented through typed, explicit, reviewable, and machine-verifiable semantics. "Language" includes the formal semantic model and does not mean unrestricted natural-language execution.
AI-assisted	AI may interpret, draft, retrieve, compare, predict, or orchestrate within an approved envelope. AI does not prescribe, approve, sign, waive QA, release, return equipment to service, or command treatment delivery.
Radiation treatment planning	The domain includes photon and proton planning and related imaging, registration, contouring, dose, robustness, review, QA, and interoperability functions within approved scope.
Model-based	Requirements, hazards, interfaces, components, policies, units, traces, verification claims, and release definitions are connected semantic models. The term is not limited to inference models.
Safety-constrained orchestration	The implementation coordinates commissioned systems through deterministic eligibility decisions, least-authority execution capabilities, reconciliation, provenance, and fail-closed behavior. It does not replace professional or commissioned-system authority.

4 Architecture interpretation

GCPL names the governed semantic layer that defines what a request can mean, what information it must contain, which uncertainties and conflicts remain, which actors may request or decide an action, which evidence and commissioned profile apply, and which artifacts may be generated.

NL-TPS names the implementation that accepts typed or natural-language input, produces candidate PlanIntent, applies deterministic kernel decisions, obtains human ProfessionalDecision records where required, issues scoped ExecutionCapability tokens, orchestrates commissioned services, and preserves provenance and audit evidence.

The naming decision does not change the four trust zones, ARCH-INVARIANT-001, the separation of PlanIntent from ProfessionalDecision, the mechanism/policy boundary, or the authority of commissioned systems and qualified people.

5 Identifier and compatibility policy

No bulk rename shall be performed merely to reflect this decision. The following remain stable unless a separate impact-assessed change is approved:

- requirement, interface, component, risk, trade, V&V, team, architecture, decision, and release identifiers;
- filenames beginning with NL_TPS;
- repository name and existing links;

- namespaces beginning with `nltps`;
- schema identifiers, API names, audit fields, and database values already released or under controlled review;
- citations to the historical ConOps and requirements baselines.

New human-readable covers and summaries may lead with GCPL while carrying the NL-TPS implementation alias. New machine identifiers shall use the existing `nltps` namespace until a separate compatibility decision defines aliases, migration, deprecation, and rollback. An identifier rename is a configuration change, not an editorial correction.

6 Communication rules

1. On first use in a document, write “Governed Clinical Planning Language (GCPL)” and identify NL-TPS as its current implementation where relevant.
2. Use the preferred descriptive title when explaining the complete program to clinical, engineering, governance, academic, or external audiences.
3. Do not describe GCPL or NL-TPS as an autonomous TPS, replacement for professional judgment, dose authority, or treatment-delivery controller.
4. Do not imply that “governed,” “clinical,” “model-based,” or “safety-constrained” constitutes regulatory clearance, accreditation, commissioning, validation, or proof of safety.
5. Preserve exact historical titles in citations and trace records. Add the GCPL alias outside the quoted title when clarification is needed.
6. Use “radiotherapy” and “radiation treatment planning” according to audience and publication style; neither changes the controlled domain scope.

7 Propagation plan

Phase	Artifacts	Action	Acceptance
1	ADR-002, terminology registry, architecture baseline, suite cover, and README	Record hierarchy and introduce GCPL / NL-TPS combined identity	Exact names agree; no identifier change; documents compile
2	New controlled documents, diagrams, schemas, UI copy, and repository guidance	Apply naming rules prospectively and maintain alias metadata	Terminology lint and review; trace links preserved
3	Existing Documents I–XVII	Revise only under normal document change control when substantive updates occur	Change record, semantic diff, citations preserved, no accidental requirement change

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Phase	Artifacts	Action	Acceptance
4	External publications, dissertation, grants, contracts, and regulatory records	Select the controlled program or research title appropriate to the venue	Sponsor, legal/regulatory, academic, communications, and authorship review as applicable

This decision does not authorize retrospective automated replacement across the corpus. Such replacement could alter exact normative text, citations, source hashes, generated mirrors, schemas, tests, or evidence links.

8 Alternatives considered

Alternative	Benefit	Limitation	Disposition
Retain NL-TPS as the sole name	Maximum continuity	Overemphasizes interface modality and can imply a replacement TPS or chatbot	Rejected as the sole program identity
Rename everything to GCPL immediately	Uniform visible identity	Breaks stable references, hashes, repository paths, namespaces, and controlled history	Rejected
Use only the long descriptive title	High explanatory accuracy	Impractical for implementation references and identifiers	Retained as the preferred descriptive title, not the sole short name
Use GCPL for technology and NL-TPS for implementation	Accurate conceptual hierarchy with backward compatibility	Requires disciplined first-use definitions and alias management	Selected

9 Risks and mitigations

Risk	Failure condition	Mitigation and evidence	Effect
ADR2-R-01	GCPL and NL-TPS are treated as competing systems	Controlled hierarchy, first-use definition, terminology registry, and architecture cross-reference	Correct before release
ADR2-R-02	Bulk renaming breaks traceability or hashes	Stable identifier policy; prospective propagation; semantic diff and import regeneration only through change control	Blocks rename

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Risk	Failure condition	Mitigation and evidence	Effect
ADR2-R-03	“Clinical” or “safety” is interpreted as an approval claim	Mandatory non-authorization notice and communications review	Correct claim
ADR2-R-04	“AI-assisted” is interpreted as autonomous authority	Preserve A0–A4 boundaries, professional-decision separation, and explicit prohibited-action language	Blocks misleading use
ADR2-R-05	“Model-based” is confused with inference models only	Define MBSE/semantic-model meaning and separately identify inference-model lifecycle	Correct terminology
ADR2-R-06	Acronym proliferation creates inconsistent citations	Reserve GCPL for Governed Clinical Planning Language; retain NL-TPS as alias; prohibit a competing acronym without decision	Blocks new acronym
ADR2-R-07	External title changes obscure historical baselines	Preserve exact source titles and versioned alias metadata in citations and manifests	Requires correction

10 Verification and acceptance

ADR-002 is implemented for the current documentation increment when:

1. the preferred descriptive title, subtitle, GCPL technology name, NL-TPS implementation alias, and research title exactly match this decision;
2. the machine-readable terminology and architecture records carry the same hierarchy;
3. the architecture baseline identifies GCPL and NL-TPS without changing the product boundary or authority model;
4. the controlled suite lists all constituent documents, includes ADR-002, and retains the legacy file and identifier structure;
5. terminology and trace checks find no accidental identifier rename or unsupported authority claim;
6. the updated standalone documents and combined review suite compile and render without new fatal or unresolved-reference errors.

Approval of a dissertation, paper title, trademark, external brand, regulatory name, contract term, or clinical product label remains outside this ADR and requires the applicable authority.

11 Rollback and supersession

Before external adoption, rollback consists of returning covers to the NL-TPS-only presentation while retaining ADR-002 as a superseded decision record. After external adoption, a replacement decision shall define alias continuity, affected publications and agreements, identifier impact, redirect policy, and correction responsibilities.

Any successor title shall preserve the product boundary: AI assists; deterministic policy decides eligibility; commissioned systems perform their assigned calculations and records functions; qualified people retain clinical authority.

Approval record

Accountability	Name	Decision/date	Conditions or dissent
Program sponsor	TBD	Not recorded	Program identity and external-use approval
Architecture board	TBD	Not recorded	Terminology and architecture consistency
Clinical governance	TBD	Not recorded	Clinical interpretation and authority boundary
Quality/configuration management	TBD	Not recorded	Controlled propagation and trace preservation
Regulatory, legal, and communications	TBD	Not recorded	Required for applicable external claims and use